

JAWAD UL HADI

Backend Lead / Architect — AI-First Systems Design

NestJS · TypeScript · PostgreSQL · GraphQL · Microservices

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PROFESSIONAL SUMMARY

Backend Lead / Architect with 7+ years designing and independently owning scalable, multi-tenant SaaS platforms and AI-powered enterprise systems — architecture, data model, and reliability strategy included. Track record: 90%+ reduction in query latency, 60% cut in AI integration complexity, 99.9% uptime across production systems serving enterprise-scale customers. Operates AI-first — using Claude Code and GitHub Copilot for scaffolding and implementation, Gemini 2.5 Flash for parsing and conversational features, and Jest for test coverage — while holding full ownership of correctness, security, and system design. Author of “Designing for AI Failure,” a technical case study on production resilience architecture. Based in Islamabad, Pakistan; schedule fully aligned to EST/PST hours.

CORE SKILLS

- **Architecture & Systems Design:** Microservices, GraphQL API design, event-driven architecture, multi-tenant SaaS, technical direction, scalability/reliability trade-offs
- **Backend Engineering:** Node.js, NestJS, TypeScript, Python, FastAPI, REST/GraphQL, PostgreSQL, MongoDB, Redis, query optimization
- **AI-First Engineering:** Claude Code, GitHub Copilot, Gemini 2.5 Flash, AI agent workflows for scaffolding/implementation/testing, RAG pipelines, LLM provider integration (OpenAI, Gemini, Anthropic)
- **Reliability & Delivery:** CI/CD (GitHub Actions), Jest, automated testing strategy, OAuth 2.0/JWT security, AWS/GCP, Docker/Kubernetes

PROFESSIONAL EXPERIENCE

Backend Lead / Senior Software Engineer | MicroAgility Services (Pvt) Ltd.

Jan 2024 – Aug 2026

- Cut AI integration complexity by 60% and eliminated user-facing failures during provider outages by architecting a provider-agnostic microservices layer that unified OpenAI, Gemini, and Anthropic behind a single abstraction with automatic failover.
- Achieved zero user-facing AI failures during live LLM outages, measured across all production incidents by designing a three-tier resilience pattern (Retry → RAG Fallback → Rule-Based Floor) and defining it as the team's standard failure-handling architecture.
- Reduced dashboard response times from 12s to under 2s (83% improvement) by owning end-to-end PostgreSQL and MongoDB query optimization, indexing strategy, and MeiliSearch rollout.
- Sustained sub-second API response times under thousands of concurrent async jobs by designing an event-driven processing architecture on BullMQ and Redis and setting the technical direction for how the team handles high-throughput workloads.
- Shortened release cycles and lifted developer productivity, measured by faster time-to-deploy by building GitHub Actions CI/CD pipelines with automated test coverage as a standing testing strategy rather than a per-project afterthought.
- Held zero cross-tenant data exposure incidents across the platform's lifetime by architecting secure REST and GraphQL APIs on OAuth 2.0/JWT and enforcing tenant isolation as a non-negotiable design standard.
- Delivered a complete timesheet module end-to-end, measured by full Jest unit-test coverage at handoff by using GitHub Copilot as the primary implementation tool under the same review and correctness bar as hand-written code.

- Built an internal ATS platform, measured by shipped production features by using Claude Code as an AI pair-programmer for scaffolding and implementation and Gemini 2.5 Flash for document parsing and the conversational assistant, while retaining full ownership of architecture and review.
- Raised engineering consistency across the team, measured by review-standard adoption by mentoring backend engineers through architecture and code review covering both AI-assisted and hand-written code.
- Authored “Designing for AI Failure,” a technical case study documenting the platform's three-tier resilience architecture (Retry → RAG Fallback → Rule-Based Floor) with system diagrams and a reversible vector-store ADR, used internally to align technical direction and onboard engineers.

Backend Software Engineer | MicroAgility Services (Pvt) Ltd.

Feb 2022 – Jan 2024

- Maintained 99.9% uptime through zero-downtime schema migrations by designing a migration process that ran against live multi-tenant production databases.
- Improved data retrieval speed by ~40% by refactoring REST and GraphQL services serving thousands of users across multiple tenants.
- Cut manual onboarding effort for new enterprise tenants by automating the tenant's deployment and migration pipeline.

Software Engineer | Market Icon

Jan 2018 – Mar 2022

- Eliminated idle infrastructure cost by designing serverless AWS Lambda/API Gateway systems on event-driven architecture instead of always-on servers.
- Enabled secure third-party integrations (CRM, payments) by building APIs across Express.js, FastAPI, and Django REST Framework.

EDUCATION AND CERTIFICATIONS

B.S. Computer Science — Government College University, Faisalabad, 2017

Certified across AI/LLM Systems, Cloud Architecture, and Backend Engineering (IBM, Microsoft, Google, Anthropic) — full credential list in [Portfolio](#)

SELECTED TECHNICAL WRITING

“Designing for AI Failure” — case study on a retrieval-grounded, three-tier fallback architecture (Retry → RAG Fallback → Rule-Based Floor) for production NestJS systems, incl. system diagrams and a reversible vector-store ADR. [Read it in the Case Study](#)

ADDITIONAL INFORMATION

- **Languages:** English (professional working proficiency), Urdu (native)
- **Availability:** Fully remote; schedule shifted to align with EST/PST hours as required